

SCARFACE: a harmonized spatio-temporal dataset integrating socio-economic, environmental, and agricultural indicators for the Po Valley (Italy), 2011–2024

Methodological note

Zenodo Version v2, Published on April 15th, 2026

Paolo Maranzano^{1,2}, Pietro Colombo³, Felicetta Carillo⁴,
Riccardo Borgoni¹, Riccardo Pajno¹, Matteo Borrotti¹,
Luca Ferrero⁵ and Ezio Bolzacchini⁵

¹University of Milano-Bicocca, Department of Economics, Management and Statistics (DEMS),
Piazza dell’Ateneo Nuovo n.1, 20100, Milano, Italy

²Fondazione Eni Enrico Mattei, Corso Magenta n.63, 20123, Milano, Italy

³School of Mathematics and Statistics, University of Glasgow, UK

⁴Italian Council for Agricultural Research and Economics –

Research Centre for Agricultural Policies and Bioeconomy (CREA-PB),

⁵University of Milano-Bicocca, Department of Earth and Environmental Sciences (DISAT)

Corresponding author: paolo.maranzano@unimib.it

April 14th, 2026

Abstract

We present *Sequestering CARbon through Forests, AgriCulture, and land use* (SCARFACE), a harmonized spatio-temporal dataset that integrates climate, air quality, airborne pollutant emissions, land cover, soil properties, agro-industry dynamics and socio-economic indicators, to jointly investigate interconnected processes linking agricultural systems, atmospheric dynamics, emissions, and socioeconomic conditions in the Po Valley, Northern Italy. The spatial reference unit is the Agrarian Sub Region, that is, groups of contiguous municipalities that are considered relatively homogeneous with respect to natural conditions, agronomic characteristics, and agricultural production systems. The dataset adopts an annual panel structure from 2011 to 2024 defined over the 256 ASRs partitioning the Po Valley and comprises more than 2,500 indicators sourced from national and international public institutions. Heterogeneous data are harmonized within a processing workflow, tailored to the specific characteristics of each dataset, that guarantee spatial and temporal consistency of the output dataset. The resource supports reuse in applied econometrics, spatio-temporal modeling, clustering, and policy analysis focused on

agriculture, air quality, and land use in a major European hotspot.

Keywords: High-dimensional spatio-temporal dataset; Po Valley in Northern Italy; Agrarian regions; Agricultural activities; Air quality and pollution emissions; Meteorology; Spatial block kriging; Land cover; Livestock; Socio-economic indicators.

S1 Spatial alignment of gridded datasets via spatial block kriging

Several environmental datasets included in SCARFACE, such as meteorological variables, atmospheric concentrations, and emission inventories, are originally provided on regular spatial grids and need to be realigned to the ASR spatial support to be consistent with the final structure of the dataset.

The need to integrate information collected on different spatial supports is known in spatial statistics as the *Change of Support Problem*, extensively discussed in Gotway and Young (2002). In our case, the original data consist of point or grid-cell observations, whereas the target quantities correspond to averages over irregular polygonal domains. When the objective is to obtain spatial averages over areas, spatial block kriging (BK) represents a natural and statistically principled solution (Chiles and Delfiner, 2012).

Let $Y = \{Y(s), s \in D\}$ denote a real-valued spatial random field defined over the geographical region D , and let $V \subset D$ be a polygonal subregion corresponding to one ASR with area $|V|$. Block kriging is used to predict the spatial average of the random field over a target polygon V , defined as

$$Y_V = \frac{1}{|V|} \int_V Y(s) ds. \quad (1)$$

The ordinary block kriging predictor is obtained as the best linear unbiased predictor (BLUP) of the spatial average Y_V using a linear combination of observations at nearby sampling locations:

$$\hat{Y}_V = \sum_{i=1}^k \alpha_i Y(s_i) = \boldsymbol{\alpha}^\top \mathbf{Y},$$

where $\mathbf{Y} = (Y(s_1), \dots, Y(s_k))^\top$ is the vector of observations (i.e., values at grid-cell centers) selected in the local neighborhood and $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_k)^\top$ is the vector of kriging weights.

The vector of kriging weights is obtained as the solution to the constrained optimization problem

$$(\alpha_1, \dots, \alpha_k) = \arg \min_{\alpha_1, \dots, \alpha_k} \text{Var} \left(Y_V - \sum_{i=1}^k \alpha_i Y(s_i) \right) \quad (2)$$

subject to the unbiasedness constraint

$$\sum_{i=1}^k \alpha_i = 1. \quad (3)$$

This optimization depends on the spatial covariance structure of the process. In point kriging, the covariance vector is composed of point-to-point covariances between each observation location s_i and the prediction location u . In block kriging, the relevant covariance terms are defined with respect to the target block support rather than to a single prediction location, and are evaluated through block-to-point and block-to-block covariance integrals, which in practice are approximated numerically using discretization

of the polygonal support (detailed derivations can be found in Chiles and Delfiner, 2012). Specifically, if the block V is discretized by N_V representative points $u_1^{(V)}, \dots, u_{N_V}^{(V)}$, the point-to-block covariance between an observation location s_i and the target block V is approximated as the arithmetic mean of the point-to-point covariances between s_i and all discretization points of the block:

$$C(s_i, V) \approx \frac{1}{N_V} \sum_{j=1}^{N_V} C(s_i, u_j^{(V)}). \quad (4)$$

Hence, $C(s_i, V)$ measures how strongly the observation at location s_i co-varies with the *average value* of the process over the whole support V , rather than with the process at a single location. Geometrically, this quantity is obtained by connecting the point s_i to every discretization point inside the target block and averaging the corresponding point-to-point covariances.

More generally, if two blocks V and V' are discretized by $\{u_1^{(V)}, \dots, u_{N_V}^{(V)}\}$ and $\{u_1^{(V')}, \dots, u_{N_{V'}}^{(V')}\}$, respectively, the block-to-block covariance is approximated by the average of all pairwise point-to-point covariances between discretization points in the two blocks:

$$C(V, V') \approx \frac{1}{N_V N_{V'}} \sum_{i=1}^{N_V} \sum_{j=1}^{N_{V'}} C(u_i^{(V)}, u_j^{(V')}). \quad (5)$$

This quantity expresses the covariance between the block averages over V and V' . As a special case, when $V' = V$, one obtains the block-to-block covariance

$$C(V, V) \approx \frac{1}{N_V^2} \sum_{i=1}^{N_V} \sum_{j=1}^{N_V} C(u_i^{(V)}, u_j^{(V)}), \quad (6)$$

which corresponds to the covariance of the block average with itself and enters the expression of the block kriging variance.

The above optimization problem can also be recast into the following ordinary kriging system

$$\begin{bmatrix} \mathbf{C} & \mathbf{1} \\ \mathbf{1}^\top & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{\alpha} \\ \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{c}_V \\ 1 \end{bmatrix}, \quad (7)$$

where $\mathbf{C}$ is the covariance matrix among the selected observations, $\mathbf{c}_V = (C(s_1, V), \dots, C(s_k, V))^\top$ is the point-to-block covariance vector, and λ is the Lagrange multiplier enforcing the unbiasedness constraint $\sum_{i=1}^k \alpha_i = 1$. The resulting predictor is the best linear unbiased predictor of the block mean, and the associated block kriging prediction variance is

$$\widehat{\text{Var}}(\hat{Y}_V - Y_V) = C(V, V) - \boldsymbol{\alpha}^\top \mathbf{c}_V + \lambda. \quad (8)$$

The spatial prediction is implemented using a local ordinary block kriging algorithm (Pebesma, 2004) with constant but unknown mean (i.e., ordinary kriging with intercept-only trend specification) and depending on the spatial covariance structure of the process, typically expressed through a semi-variogram function. The block kriging strategy is applied for each combination of variable, year and sector (when applicable), such that only the spatial field observed at grid-cell location and a subset of nearby observations is used in

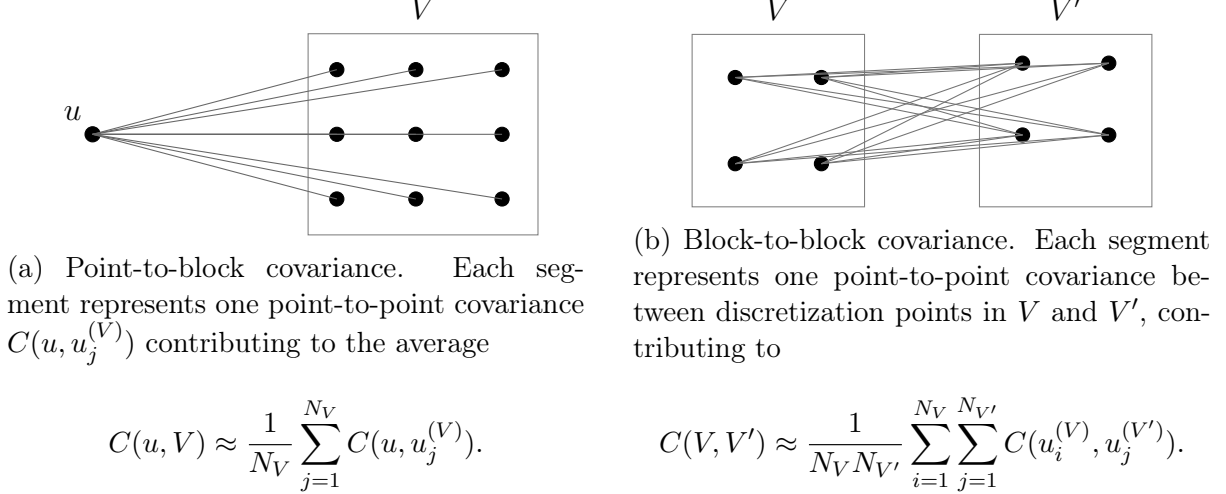


Figure 1: Schematic representation of the covariance terms involved in block kriging. Left: point-to-block covariance between a point support and a target block. Right: block-to-block covariance between two block supports.

to predict the average value over each ASR polygon. The kriging predictor is constructed as a weighted linear combination of nearby observations, where the set of contributing observations is restricted to the closest sampling locations around the target block. The locality of the prediction is controlled by the size of the kriging neighborhood (i.e., n_{max}), that is, the maximum number of neighboring observations included in the predictor. Smaller neighborhoods produce more localized predictions that reflect short-range spatial variability, whereas larger neighborhoods yield smoother predictions that capture broader spatial structures but increase computational burden. This local formulation reduces computational cost (the kriging system scales approximately as $O(n_{max}^3)$ with respect to the number of neighboring observations) and improves numerical stability when working with large gridded datasets.

In practice, the up-scaling procedure consists of three main steps. First, a spatial covariance model is estimated. Several isotropic covariance families are considered, namely spherical, exponential, Gaussian, and Matérn models. For each candidate covariance model, prediction performance is evaluated using repeated out-of-sample predictions obtained through random 5-fold cross-validation (CV) (Otto et al., 2024). The covariance model producing the lowest root mean squared prediction error (RMSE) is selected. Second, for the selected covariance model, the a grid of candidate neighborhood size is tuned through CV, and the value minimizing the RMSE is retained. Third, using the selected covariance structure and neighborhood configuration, ordinary block kriging is performed over the ASR polygons. The procedure yields both the predicted spatial mean for each ASR and the corresponding kriging prediction variance, which quantifies the interpolation uncertainty associated with the area-level spatial prediction. The spatial alignment procedure based on local block kriging is summarized in Algorithm 1.

In the implementation adopted in SCARFACE, block kriging predictions and associated uncertainties are computed using the `krige` function from the `gstat` package (Pebesma, 2004) in R. In `gstat`, the point-to-block and block-to-block covariances are approximated numerically by discretizing the target polygonal support V into a finite set of representa-

Algorithm 1 Local block kriging procedure for ASR spatial alignment

Require: Gridded observations $\{(s_i, Y(s_i)) : i = 1, \dots, n\}$ for a given variable, year, and sector; ASR polygons $\{V_d : d = 1, \dots, D\}$; candidate covariance families $\mathcal{C}$; candidate neighborhood sizes $\mathcal{N}$

Ensure: For each ASR V_d , block kriging estimate $\hat{Y}_{V_d}$ and prediction variance $\widehat{\text{Var}}(Y_{V_d} - \hat{Y}_{V_d})$

1: **Covariance-model selection**

2: Fix an initial neighborhood size $n_{\max}^{(0)}$

3: **for all** $C \in \mathcal{C}$ **do**

4: Fit the isotropic variogram model corresponding to covariance family C

5: Perform random 5-fold cross-validation using local ordinary kriging with the $n_{\max}^{(0)}$ closest observations

6: Compute the cross-validated root mean squared error $RMSE(C \mid n_{\max}^{(0)})$

7: **end for**

8: Select

$$C^* = \arg \min_{C \in \mathcal{C}} RMSE(C \mid n_{\max}^{(0)})$$

9: **Neighborhood-size tuning**

10: **for all** $n_{\max} \in \mathcal{N}$ **do**

11: Perform random 5-fold cross-validation using covariance family C^*

12: Compute the cross-validated root mean squared error $RMSE(n_{\max} \mid C^*)$

13: **end for**

14: Select

$$n_{\max}^* = \arg \min_{n_{\max} \in \mathcal{N}} RMSE(n_{\max} \mid C^*)$$

15: **Block kriging over ASR polygons**

16: **for** $d = 1, \dots, D$ **do**

17: Use covariance family C^* and neighborhood size $n_{\max}^*$

18: Perform local ordinary block kriging to predict the block mean Y_{V_d}

19: Compute the predicted block mean $\hat{Y}_{V_d}$

20: Compute the block kriging prediction variance $\widehat{\text{Var}}(Y_{V_d} - \hat{Y}_{V_d})$

21: **end for**

22: **return** $\{\hat{Y}_{V_d}, \widehat{\text{Var}}(Y_{V_d} - \hat{Y}_{V_d}) : d = 1, \dots, D\}$

tive points and averaging the implied point-to-point covariances. When prediction targets are supplied as polygons, the package automatically computes the block average for each polygon and returns both the block kriging prediction and the corresponding prediction variance. In our implementation, prediction is further localized through the parameter `nmax`, which restricts the kriging system to the closest observations in the neighborhood, and the intercept-only specification `1` is used throughout the fitting and prediction steps (Pebesma, 2004).

S2 Post-stratification procedure adopted to generate the spatio-temporal weighting system for FADN survey data

This appendix provides a detailed description of the calibration procedure used to construct the spatio-temporal weighting system linking the FADN sample to the reconstructed population of farms at the ASR level. The procedure combines direct post-stratification via iterative proportional fitting (IPF) or raking (Kolenikov, 2014; Lomax and Norman, 2016) with a hierarchical calibration strategy designed to address sparse sample coverage across strata.

The final weighting system is obtained by applying the following hierarchy within each ASR–year domain:

1. direct cell post-stratification when all strata are observed;
2. two-dimensional raking calibration on both margins;
3. one-dimensional calibration on a single margin;
4. temporal donor reconstruction using nearby years;
5. uniform domain weighting as a final fallback.

This hierarchical strategy ensures that the resulting weights remain stable and well defined even under sparse sampling conditions while preserving consistency with the reconstructed population totals at the ASR level.

S2.1 Notation

Let $d = 1, \dots, D$ denote the ASRs, $s \in \{1, 2\}$ the economic size classes (i.e., small, large), $t \in \{1, 2, 3\}$ the technical specialization classes (i.e., crop, livestock, mixed), and y the year.

Let N_{dsty} denote the number of farms in the population and n_{dsty} the number of sampled farms in the FADN survey for the corresponding stratum.

Population margins are defined as

$$N_{dsy} = \sum_t N_{dsty}, \quad N_{dty} = \sum_s N_{dsty}.$$

Similarly, sample margins are

$$n_{dsy} = \sum_t n_{dsty}, \quad n_{dty} = \sum_s n_{dsty}.$$

A schematic representation of the post-stratification workflow is reported in Figure 2.

The objective is to compute weights w_{dsty} such that the weighted sample reproduces the known population margins while remaining well defined even when some strata are not represented in the survey sample.

d=1

	Crop	Lives.	Mixed
Large	N_{111}	N_{112}	N_{113}
Small	N_{121}	N_{122}	N_{123}

d

	Crop	Lives.	Mixed
Large	N_{d11}	N_{d12}	N_{d13}
Small	N_{d21}	N_{d22}	N_{d23}

d=D

	Crop	Lives.	Mixed
Large	$N_{D1,1}$	N_{D12}	N_{D13}
Small	$N_{D2,1}$	N_{D22}	N_{D23}

Census

...

...

	Crop	Lives.	Mixed
Large	n_{111}	n_{112}	n_{113}
Small	n_{121}	n_{122}	n_{123}

d

	Crop	Lives.	Mixed
Large	n_{d11}	n_{d12}	n_{d13}
Small	n_{d21}	n_{d22}	n_{d23}

d=D

	Crop	Lives.	Mixed
Large	n_{D11}	n_{D12}	n_{D13}
Small	n_{D21}	n_{D22}	n_{D23}

Survey

...

...

where $d=1,2,...,D$ ASRs

$t = 1,2,3$ type of farming

$s = 1,2$ economic size

Figure 2: Census-based post-stratification quantities used to construct the spatio-temporal weighting system for the FADN sample. Annual population counts of farms by economic size and technical specialization (N_{dsty}) are reconstructed at the ASR level using the 2010 and 2020 Italian Agricultural Censuses and interpolated for intermediate years. These reconstructed population totals are matched with the FADN sample counts (n_{dsty}) to derive post-stratification weights through the iterative proportional fitting procedure.

S2.2 Direct post-stratification

When the survey sample covers all strata (s, t) within a given ASR–year domain (d, y), weights are computed directly through post-stratification:

$$w_{dsty} = \frac{N_{dsty}}{n_{dsty}}.$$

This corresponds to the classical cell-weighting estimator and guarantees exact reproduction of the population counts within each stratum.

S2.3 Two-dimensional raking calibration

When some strata are missing from the sample but the support of the sample allows identification of both margins, weights are calibrated through IPF. The calibration weights are obtained by solving the system

$$\sum_t w_{dsty} n_{dsty} = N_{dsy}, \quad \sum_s w_{dsty} n_{dsty} = N_{dty}.$$

Starting from initial design weights $w_{dsty}^{(0)}$, the raking algorithm iteratively rescales the weights along the two margins until convergence is achieved. The resulting weights reproduce the population totals by economic size and by technical specialization within each ASR–year domain.

S2.4 One-dimensional calibration

If two-dimensional calibration cannot be performed because the sample lacks support for one of the margins (for example, all sampled farms belong to the same specialization class), calibration is attempted along a single margin.

Two alternative calibrations are considered:

1. calibration on economic size:

$$\sum_t w_{dsty} n_{dsty} = N_{dsy},$$

2. calibration on technical specialization:

$$\sum_s w_{dsty} n_{dsty} = N_{dty}.$$

When both calibrations are feasible, the solution minimizing the maximum absolute deviation between weighted sample counts and reconstructed population totals is selected.

S2.5 Temporal donor reconstruction

In some ASR–year domains the survey sample is too sparse to support calibration. In these cases, information from adjacent years within the same ASR is used. A temporal donor year y^* is selected according to the following criteria:

1. minimum temporal distance, that is, $|y - y^*|$,
2. best calibration quality in the donor year,
3. most recent donor year when ties occur.

Let

$$p_{dsty^*} = \frac{N_{dsty^*}}{\sum_{s,t} N_{dsty^*}}$$

denote the population share of each stratum in the donor year. The reconstructed population totals for the target year are obtained by applying these shares to the total number of farms in the target domain:

$$\tilde{N}_{dsty} = p_{dsty^*} N_{dy}.$$

When sampled farms are present in the corresponding strata, weights are derived as

$$w_{dsty} = \frac{\tilde{N}_{dsty}}{n_{dsty}}.$$

S2.6 Uniform weighting

As a final fallback, when neither calibration nor temporal reconstruction is feasible, a uniform domain weight is applied:

$$w_{dsty} = \frac{N_{dy}}{n_{dy}},$$

where

$$N_{dy} = \sum_{s,t} N_{dsty}, \quad n_{dy} = \sum_{s,t} n_{dsty}.$$

This ensures that the weighted sample reproduces the total number of farms within the ASR–year domain even when the internal stratification cannot be recovered.

S2.7 Area-level estimation of the total and the mean

Using the calibrated weights, farm-level indicators from FADN are then aggregated to the ASR–year level through design-based Horvitz and Thompson (1952) estimators (HT). For a generic farm-level variable X_{dstyk} observed for farm k belonging to stratum (d, s, t, y) , the HT estimator of the ASR total for year y and area d is

$$\hat{T}_{dy}(X) = \sum_{s=1}^2 \sum_{t=1}^3 \sum_{k=1}^{n_{dsty}} w_{dsty} X_{dstyk}.$$

The corresponding estimator of the ASR mean for area d in year y is obtained as

$$\hat{\mu}_{dy}(X) = \frac{\hat{T}_{dy}(X)}{N_{dy}}, \quad N_{dy} = \sum_{s=1}^2 \sum_{t=1}^3 N_{dsty}.$$

S3 Generalized Variance Function (GVF) methodology adopted to regularize direct estimates of the variance for FADN survey data

This appendix describes the *Generalized Variance Function* (GVF) approach adopted to regularize the sampling variances of the Horvitz and Thompson (1952) estimators (HT). The objective of the GVF procedure is to stabilize highly variable direct variance estimates, particularly in domains with small sample sizes, while preserving the design-based variance structure implied by the survey design (especially in well-sampled domains). Regularization is achieved by exploiting the systematic relationship between sampling variability and domain sample size. This allows for consistent uncertainty quantification across all domain–time observations.

S3.1 Model specification

Let $\hat{Y}_d$ denote the HT estimator for domain d , and let $\hat{V}_d$ be the corresponding direct variance estimate. The variance structure is modeled using a GVF, defined on a logarithmic scale.

Two alternative formulations are considered:

Variance formulation

$$\log(\hat{V}_d) = \alpha + \beta \log(\hat{Y}_d) + g(p_d) + u_{s(d)} + v_{t(d)} + \varepsilon_d \quad (9)$$

Relative variance formulation

$$\log(CV_d^2) = \alpha + g(p_d) + u_{s(d)} + v_{t(d)} + \varepsilon_d \quad (10)$$

where

$$CV_d^2 = \frac{\hat{V}_d}{\hat{Y}_d^2} \quad (11)$$

The first formulation models the variance directly, while the second focuses on the relative precision of the estimator. These complementary specifications allow the model to capture both scale-dependent and scale-invariant components of the variance structure.

S3.2 Precision modeling

The function $g(p_d)$ captures the relationship between sampling precision and domain-level sampling characteristics. Three alternative specifications are considered:

$$g(p_d) = \begin{cases} \gamma_1 \log(n_d) \\ \gamma_1 \log\left(\frac{n_d}{N_d}\right) \\ \gamma_1 \log(n_d) + \gamma_2 \log(N_d) \end{cases} \quad (12)$$

where n_d is the effective sample size and N_d is the domain population size.

These specifications allow the GVF to account for both absolute sample size effects and relative sampling intensity. Collinear representations are avoided by not including equivalent transformations simultaneously.

S3.3 Estimation framework

The GVF is estimated using domain-year observations with sufficiently reliable direct variance estimates. In particular, only domains satisfying

$$n_d \geq 3 \quad (13)$$

are included in the estimation sample.

Spatial and temporal heterogeneity in the variance structure are captured through random intercepts:

$$u_{s(d)} \sim N(0, \sigma_s^2), \quad v_{t(d)} \sim N(0, \sigma_t^2) \quad (14)$$

where $s(d)$ and $t(d)$ denote the spatial domain (ASR) and the time index, respectively.

S3.4 Model selection

Several candidate GVF models are obtained by combining the alternative response formulations and precision specifications. Model selection is based on three complementary criteria reflecting the objectives of variance regularization.

Coherence with direct variances The agreement between GVF predictions and direct variance estimates is measured using the logarithmic root mean squared error (RMSE):

$$RMSE_{\log} = \sqrt{\frac{1}{m} \sum_{d=1}^m \left(\log(\hat{V}_d) - \log(\hat{V}_d^{GVF}) \right)^2} \quad (15)$$

Lower values indicate that the GVF reproduces the structure of the direct variances more closely.

Only models satisfying

$$RMSE_{\log} \leq RMSE_{\log}^{\min} (1 + \tau) \quad (16)$$

with $\tau = 0.05$ are retained for further comparison.

Reduction of extreme variances To evaluate the ability of the GVF to regularize unstable estimates, attention is restricted to the upper tail of the variance distribution, that is, preference is given to the specification that most effectively reduces excessively large variances. Let $Q_{0.75}(\hat{V})$ denote the 75th percentile of the direct variances. For domains with $\hat{V}_d \geq Q_{0.75}(\hat{V})$, the reduction factor is defined as

$$R_d = \frac{\hat{V}_d^{final}}{\hat{V}_d} \quad (17)$$

and summarized through

$$Reduction_{upper} = \text{median}(R_d) \quad (18)$$

Control of variance inflation The GVF may increase some variance estimates. This behavior is quantified by

$$Increase_share = \frac{1}{m} \sum_{d=1}^m \mathbf{1} \left(\frac{\widehat{V}_d^{final}}{\widehat{V}_d} > 1.05 \right) \quad (19)$$

This indicator measures the proportion of domains where the final variance exceeds the direct variance by more than 5%. Lower values indicate fewer substantial increases.

Selection rule The final model is selected through a hierarchical rule:

1. retain models within the $RMSE_{\log}$ tolerance band;
2. among them, select the model minimizing $Reduction_{upper}$;
3. if necessary, select the model minimizing $Increase_share$;
4. break remaining ties using $RMSE_{\log}$.

S3.5 Variance prediction

After model estimation, predicted squared coefficients of variation are obtained for all domains. The corresponding variance estimates are reconstructed as

$$\widehat{V}_d^{GVF} = \widehat{CV}_d^2 \widehat{Y}_d^2 \quad (20)$$

No bias correction is applied when transforming predictions from the logarithmic scale.

S3.6 Blending of variance estimates

The final variance estimate combines the direct estimator and the GVF prediction:

$$\widehat{V}_d^{final} = w(n_d) \widehat{V}_d^{GVF} + (1 - w(n_d)) \widehat{V}_d \quad (21)$$

where the weight depends on the domain sample size:

$$w(n_d) = \begin{cases} 1 & n_d \leq 1 \\ 0.5 & n_d = 2 \\ 0 & n_d \geq 3 \end{cases} \quad (22)$$

This implies that domains with a single observation rely entirely on the GVF prediction; domains with two observations combine the GVF prediction and the direct variance equally; and domains with three or more observations retain the direct variance estimate. This rule reflects the increasing reliability of direct variance estimates as the sample size grows.

References

- Chiles, J.-P. and Delfiner, P. (2012). *Geostatistics: Modeling Spatial Uncertainty*. Wiley, Hoboken, NJ, 2nd edition.
- Gotway, C. A. and Young, L. J. (2002). Combining incompatible spatial data. *Journal of the American Statistical Association*, 97(458):632–648.
- Horvitz, D. G. and Thompson, D. J. (1952). A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, 47(260):663–685.
- Kolenikov, S. (2014). Calibrating survey data using iterative proportional fitting (raking). *The Stata Journal*, 14(1):22–59.
- Lomax, N. and Norman, P. (2016). Estimating population attribute values in a table: “get me started in” iterative proportional fitting. *The Professional Geographer*, 68(3):451–461.
- Otto, P., Fassò, A., and Maranzano, P. (2024). A review of regularised estimation methods and cross-validation in spatiotemporal statistics. *Statistics Surveys*, 18(none):299–340, 42.
- Pebesma, E. J. (2004). Multivariable geostatistics in s: the gstat package. *Computers & geosciences*, 30(7):683–691.